Digital Logic Design

Course code: CSE-2323
Credit Hour: 3

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counter

Counters are used to count the clock pulses. The clock pulses occur at regular intervals. They are constructed with flip flops and logic gates.

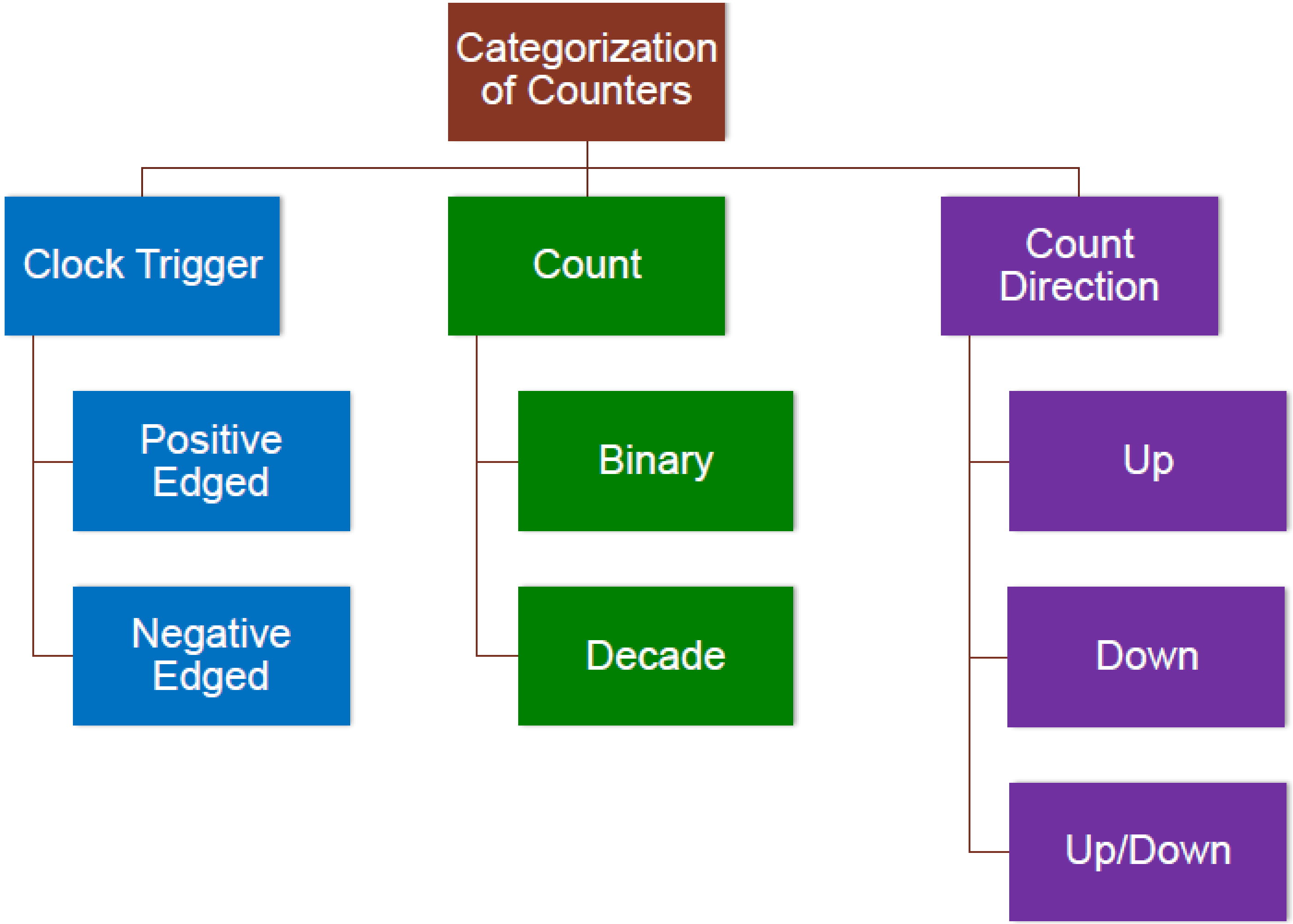
They are of two types -

1. Asynchronous counter
2. Synchronous counter

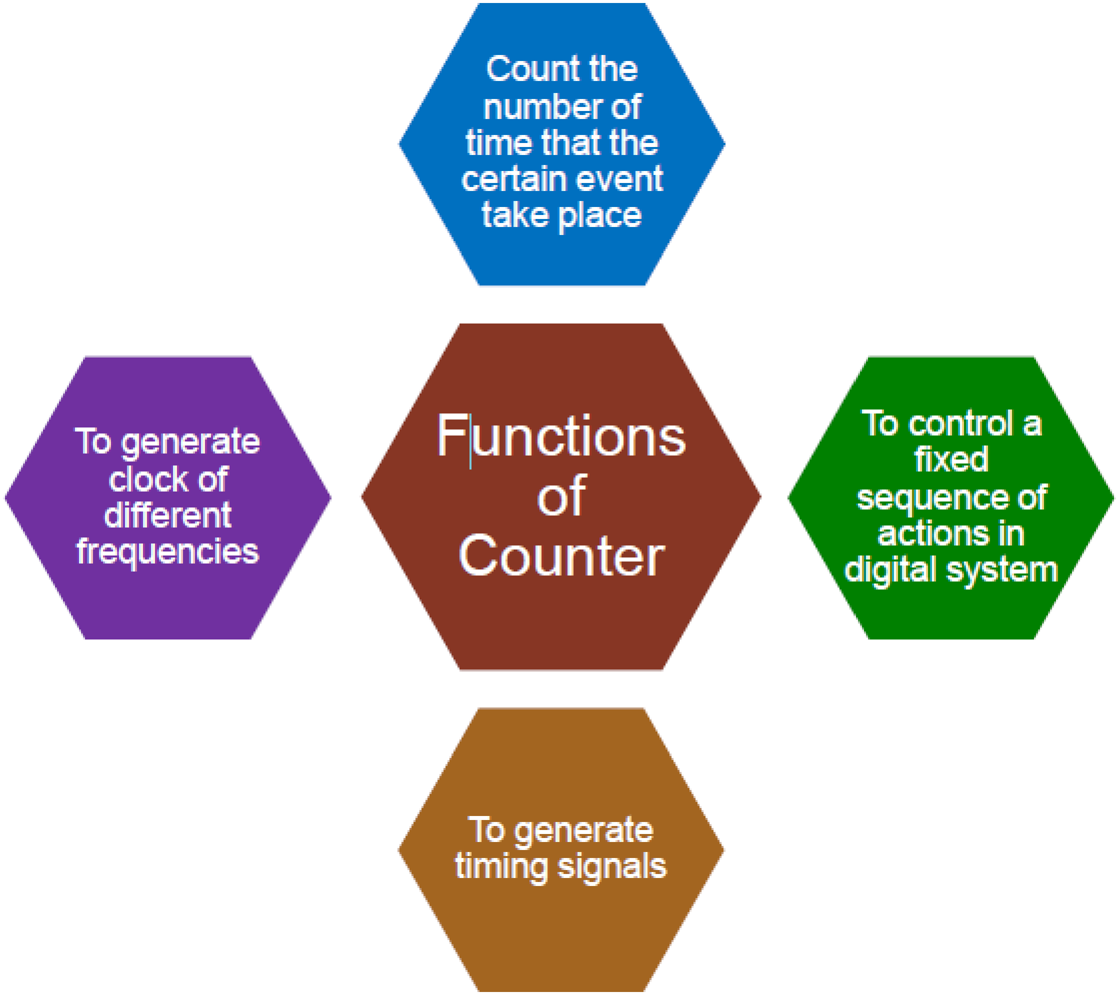
In the asynchronous counter, an external clock pulse is provided for only the first flip flop, thereafter the output of the Ist FF acts as a clock pulse for the second FF and so on.

In the case of synchronous FFs, all the flip flops are triggered simultaneously by an external clock pulse.

counter


Categorization of Counters
Clock Trigger
Positive Edged
Negative Edged
Count
Binary
Decade
Count Direction
Up
Down
Up/Down

Function Of counter



Asynchronous counter

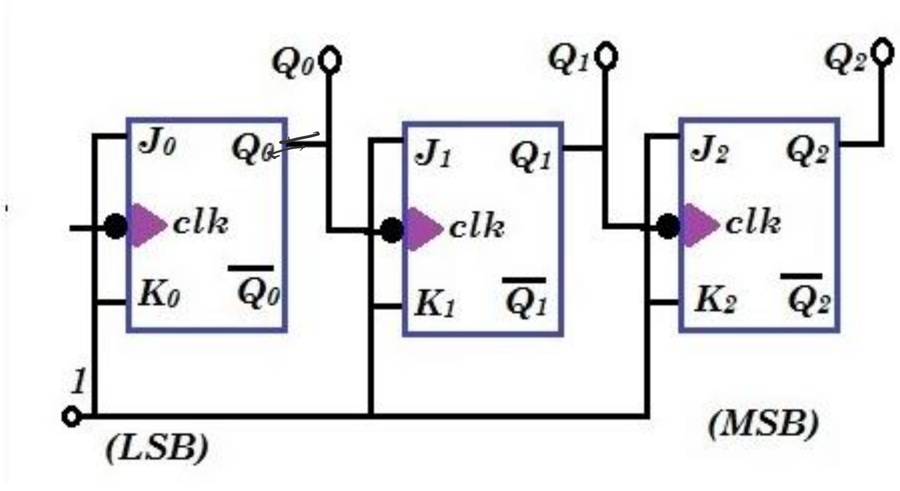
Advantages of Asynchronous counter :

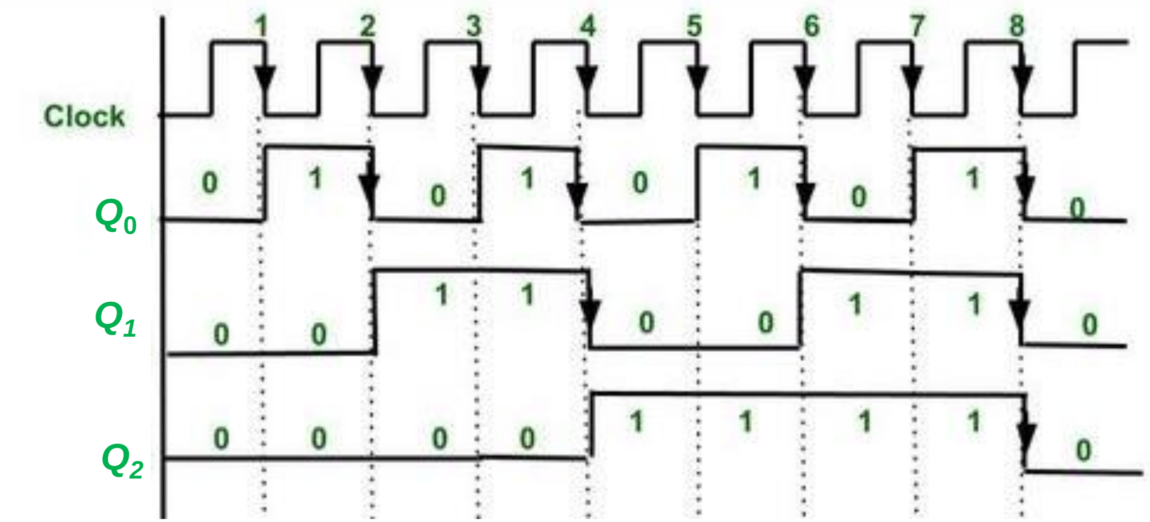
- Due to accumulation of propagation delay, they are used in low speed circuits.
- They are simple to design.
- They are used in mod n counters, and in divide by n counter that divides the input by n(i.e. n is integer).

Disadvantages of Asynchronous counter :

- As the number of flip-flops increases, the propagation delay also increases.
- For high clock frequencies, counting errors may occur because of propagation delay.

3 bit Asynchronous up counter



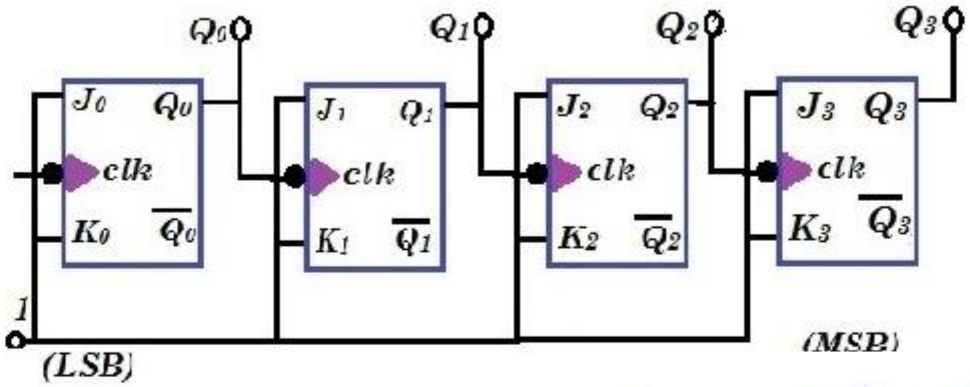


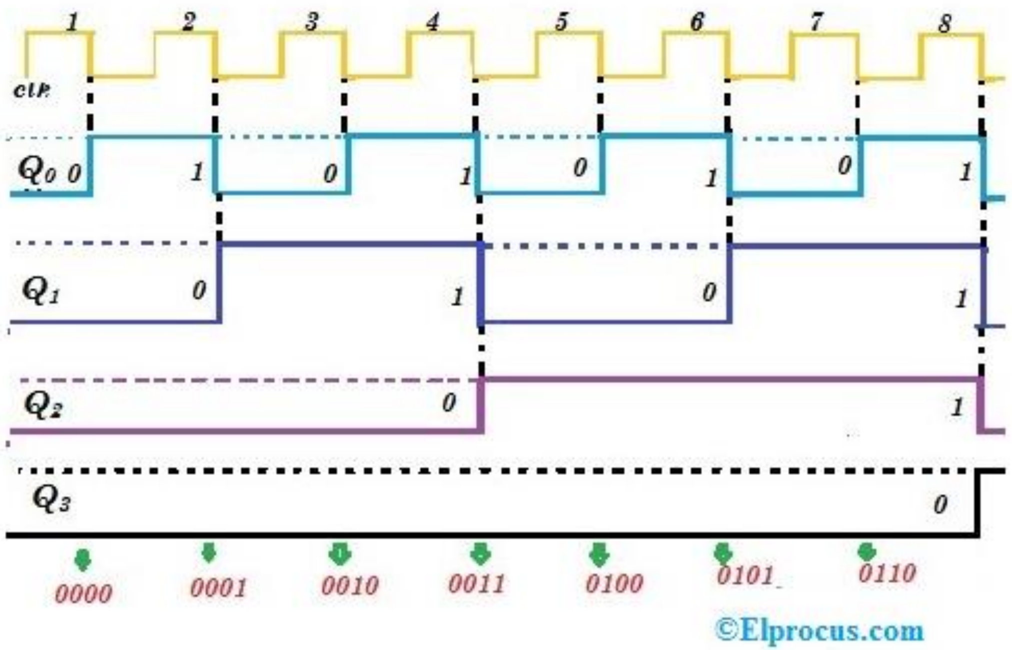
Explanation of Up counter –

- The Ist FF is connected to logic clk. Therefore, it will toggle for every falling edge.
- The 2nd FF input is connected to Q_0. Therefore it changes its state when Q_0= 1 and there is falling edge of clock.
- Similarly, 3rd FF is connected to Q_1. Therefore, it changes its state when Q_1= 1 and there is falling edge of clock.
- By this we can generate counting states of Up counter.
- After every 8th falling edge the counter is again reaching to state 0 0 0.

Therefore, it is also known as divide by 8 circuit or mod 8 counter.

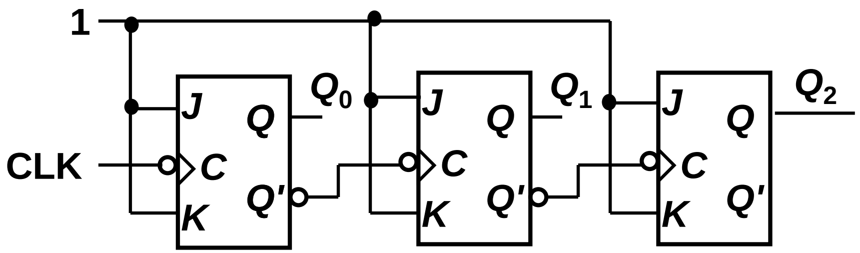
4 bit Asynchronous up counter



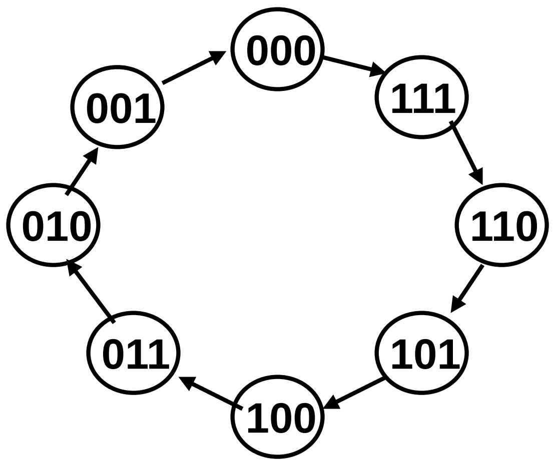


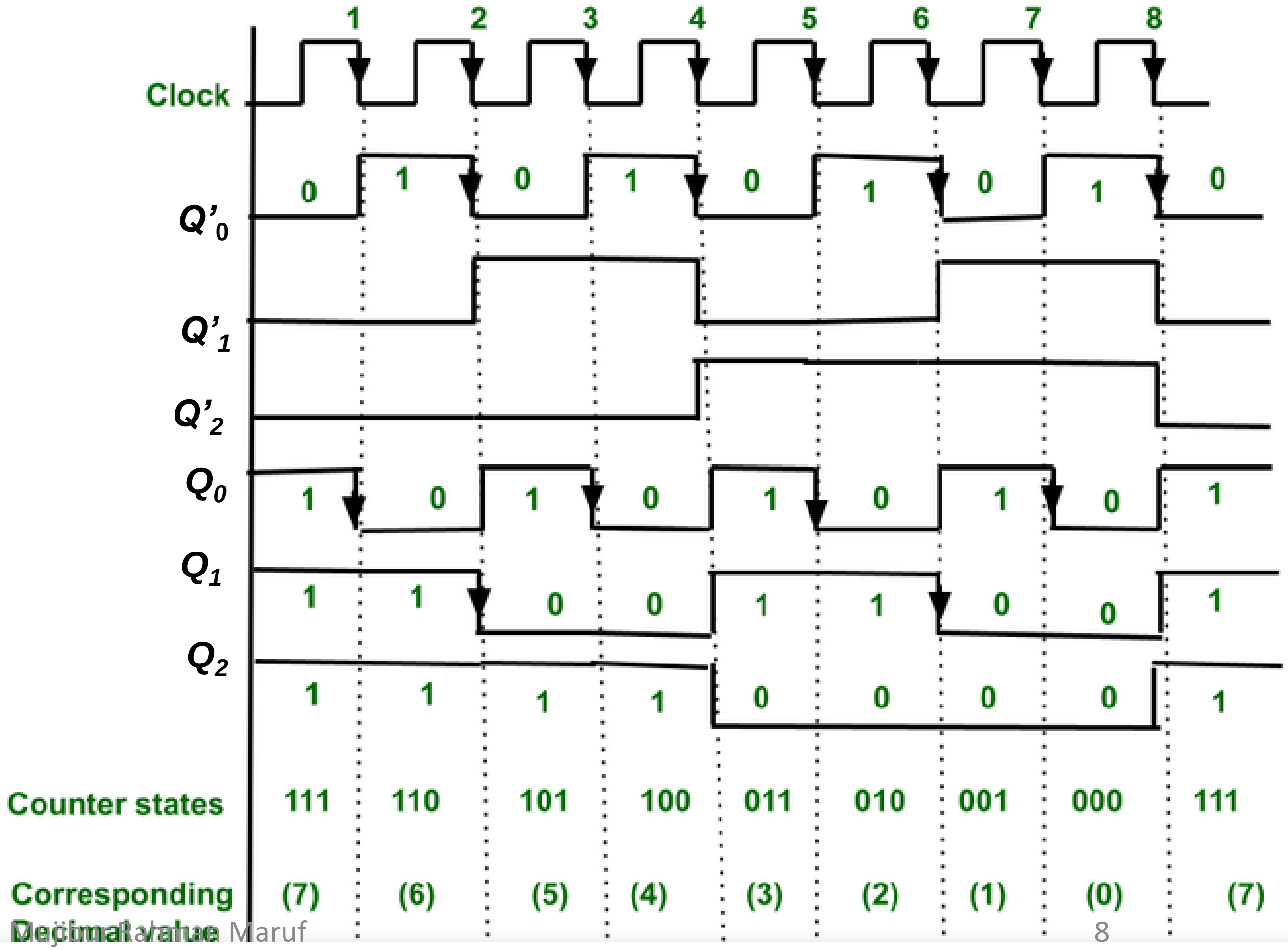
Asynchronous A 3-bit binary (MOD-8) Down Counters

A state diagram represents states with circles, and transitions between states by arrows exiting one circle and arr



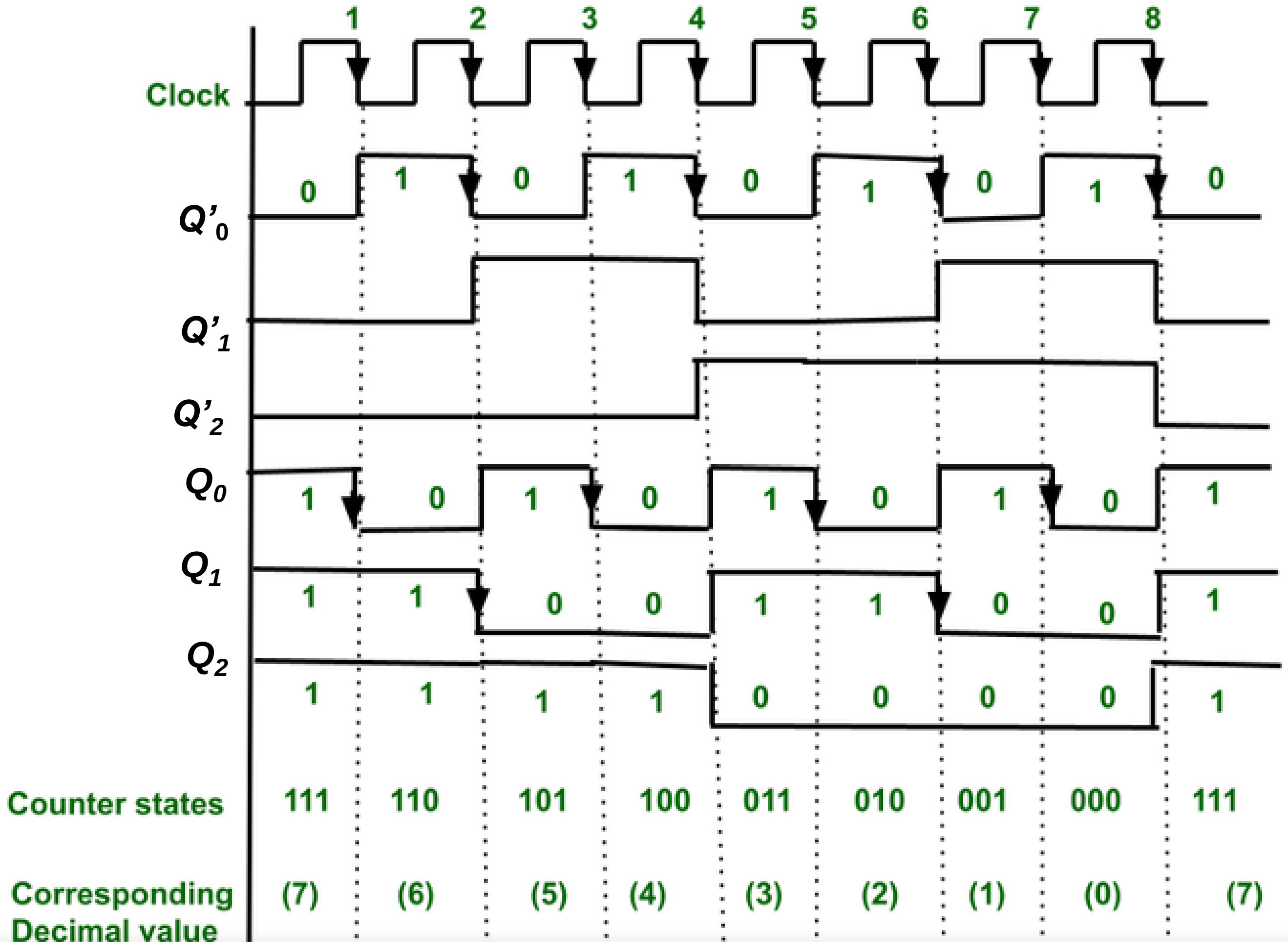
3-bit binary down counter





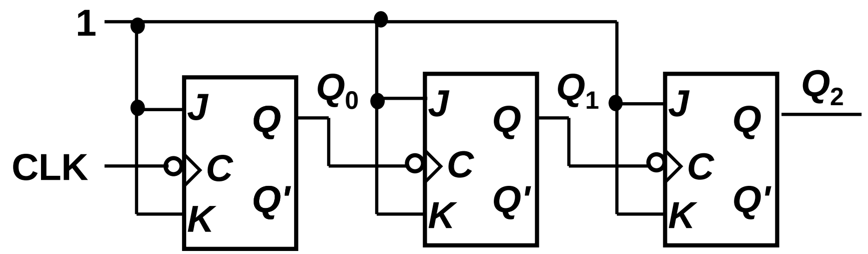
Asynchronous A 3-bit binary (MOD-8) Down Counters

- The -ve edge clock pulse is provided to Ist counter Therefore, the output state of the first counter(i.e. Q_0) will be toggled at every falling edge of the clock pu
- As the Q'_0 is feed as a clock to the second FF, therefore the output state (i.e. Q_1) will be toggled at every falling edge of Q'_0.
- In the same manner, the Q'_1 acts as clock for the third FF, therefore the output state (Q_2) of the third FF will be toggled for every falling edge of Q'
- After the 8th falling edge of the external clock puls the counter is reset to 000

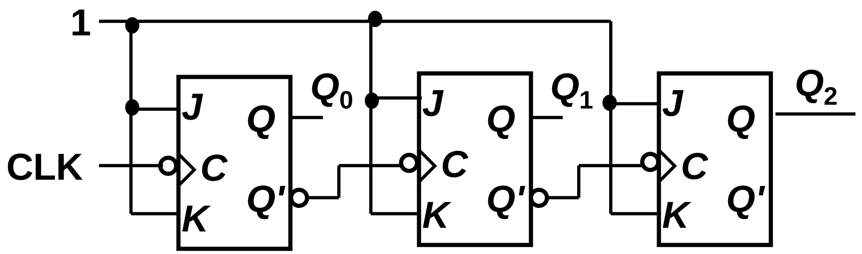


Design of 3 bit Asynchronous up/down counter

1.In this a mode control input (say M) is used for selecting up and down mode.
2.A combinational circuit is required between each pair of flip-flop to decide whether to do up or do down counting



3-bit binary up counter



3-bit binary down counter

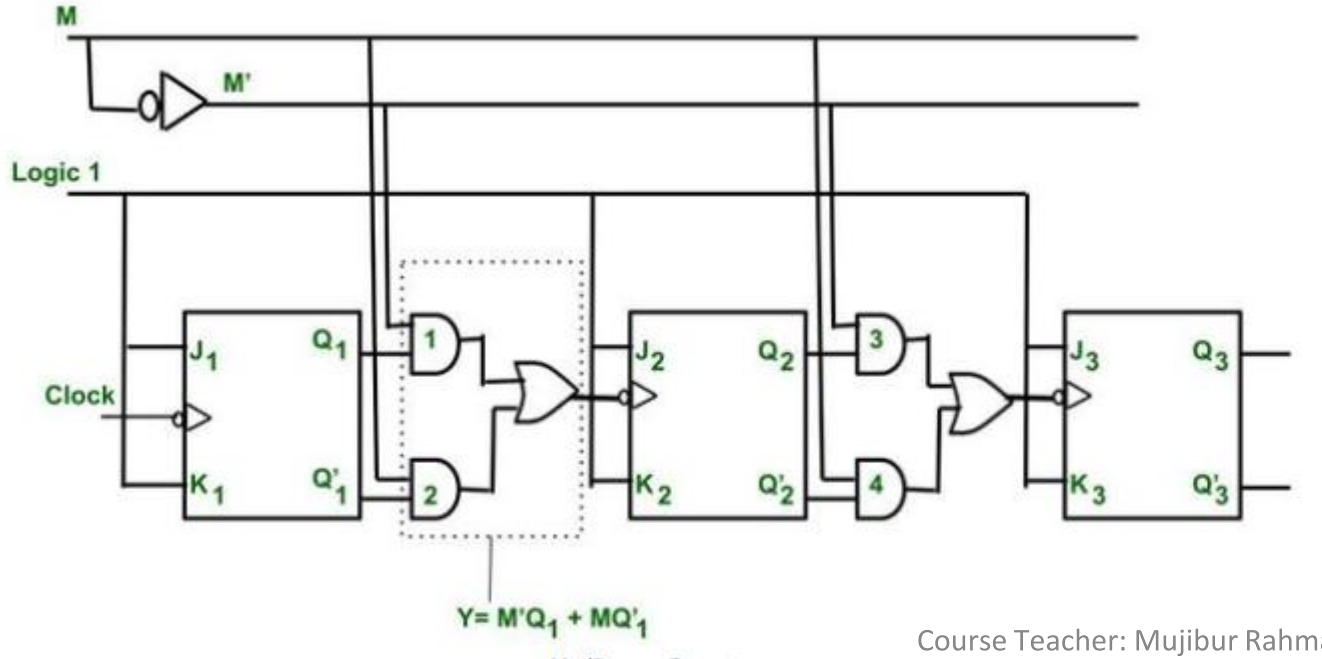
COUNT-UP Mode				COUNT-DOWN Mode			
States	Q_C	Q_B	Q_A	States	Q_C	Q_B	Q_A
0	0	0	0	7	1	1	1
1	0	0	1	6	1	1	0
2	0	1	0	5	1	0	1
3	0	1	1	4	1	0	0
4	1	0	0	3	0	1	1
5	1	0	1	2	0	1	0
6	1	1	0	1	0	0	1
7	1	1	1	0	0	0	0

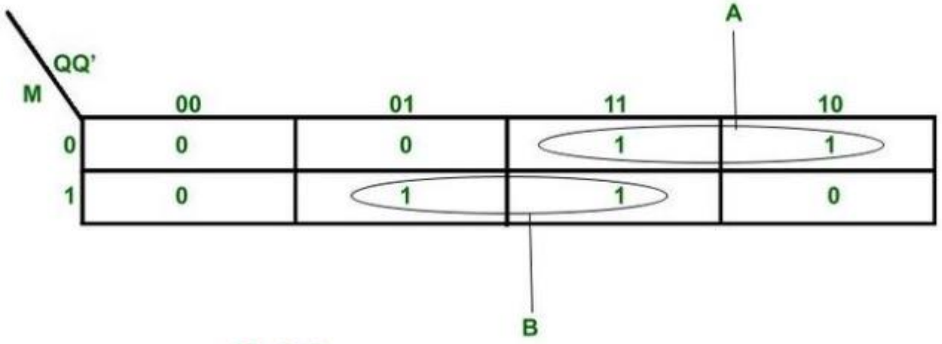
Design of 3 bit Asynchronous up/down counter

When M = 0, then Y= Q, therefore it will perform Up counting
When M = 1, then Y= Q' therefore it will perform Down counting.

Y = Q when M= 0
Y = Q' when M=1

M	Q	Q'	Y
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1





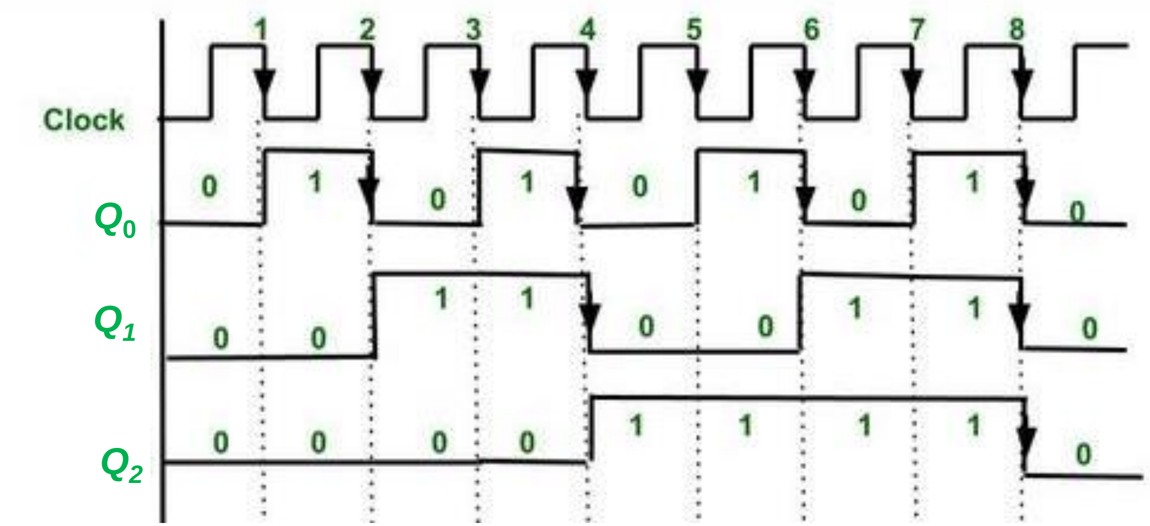
Y = A+B

Y= M'Q + MQ'

K map for finding Y

Design of 3 bit Asynchronous up/down counter

Case I – When M=0, then M' =1.
Put this in Y= **M'Q + MQ'= Q** So Q is acting as clock for next FFs.
Therefore, the counter will act as Up counter.



When M = 0 Up Counter

Explanation of Up counter –

- The 1st FF is connected to logic 1. Therefore, it will toggle for every falling edge.
- The 2nd FF input is connected to Q_0. Therefore it changes its state when Q_0= 1 and there is falling edge of clock.
- Similarly, 3rd FF is connected to Q_1. Therefore, it changes its state when Q_1= 1 and there is falling edge of clock.
- By this we can generate counting states of Up counter.
- After every 8th falling edge the counter is again reaching to state 0 0 0.

Therefore, it is also known as divide by 8 circuit or mod 8 counter.

Design of 3 bit Asynchronous up/down counter

Design of 3 bit Asynchronous up/down counter :

Case 2 – When $M=1$, then $M' = 0$.
Put this in $\boldsymbol{Y = M'Q + MQ' = Q'}$. So Q' is acting as clock for next FFs.
Therefore, the counter will act as Down counter.

Explanation of Down counter –

- The 1st FF is connected to logic 1. Therefore, it will toggle for every falling edge.
- The 2nd FF input is connected to Q'_0. Therefore it changes its state when $Q'_0 = 1$ and there is falling edge of clock.

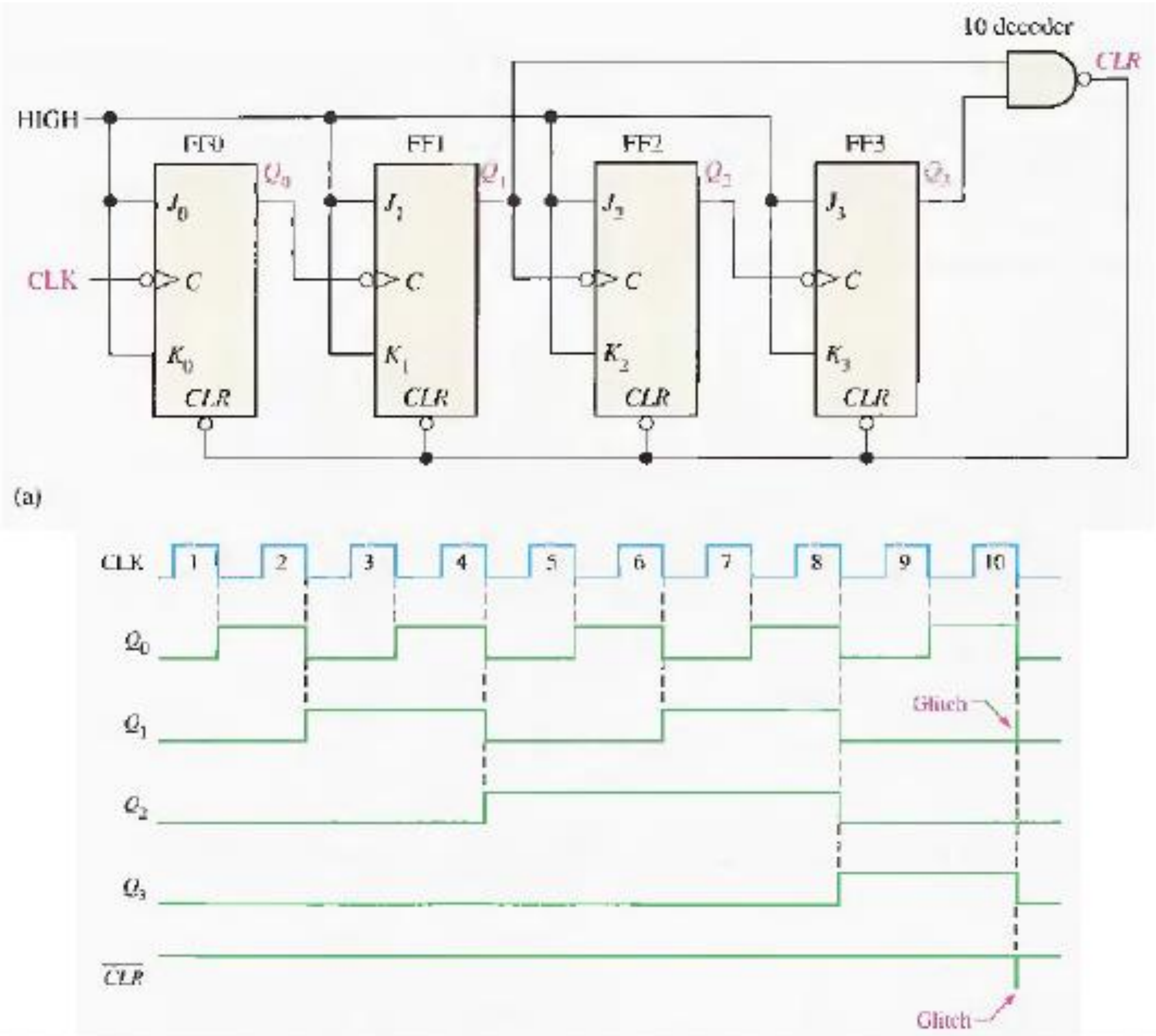
When $M = 1$ Down Counter

- Similarly, 3rd FF is connected to Q'_1. Therefore, it changes its state when $Q'_1 = 1$ and there is falling edge of clock.
- By this we can generate counting states of down counter.
- After every 8th falling edge the counter is again reaching to state 0 0 0.

Therefore, it is also known as divide by 8 circuit or mod 8 counter.

Asynchronous Decade Counters

follow the class lecture



Synchronous Counters Design procedure

Step 1: DecideThe number of flip-flops required to design counter .

Step 2: Drawing the state diagram.

Step 3: make state table

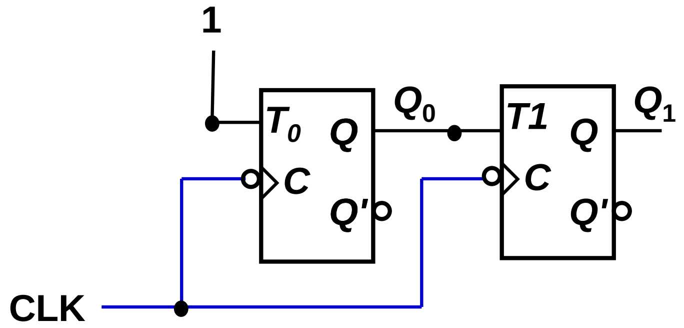
Step 4 Draw circuit

2 Bit Synchronous UP Counters

- *Example: 2-bit synchronous binary counter (using T flip-flops, or JK flip-flops with identical J,K inputs).*

Present state		Next state		Flip-flop inputs	
Q_1	Q_0	Q_1^+	Q_0^+	T_1	T_0
0	0	0	1	0	1
0	1	1	0	1	1
1	0	1	1	0	1
1	1	0	0	1	1

$T_1 = Q_0$
$T_0 = 1$



3 bit Synchronous (Parallel) Counters

Synchronous (parallel) counters: the flip-flops are clocked at the same time by a common clock pulse

3-bit synchronous binary counter (using T flip-flops, or JK flip-flops with identical J, K inputs).

Present state			Next state			Flip-flop inputs		
Q_2	Q_1	Q_0	Q_2^+	Q_1^+	Q_0^+	T_2	T_1	T_0
0	0	0	0	0	1	0	0	1
0	0	1	0	1	0	0	1	1
0	1	0	0	1	1	0	0	1
0	1	1	1	0	0	1	1	1
1	0	0	1	0	1	0	0	1
1	0	1	1	1	0	0	1	1
1	1	0	1	1	1	0	0	1
1	1	1	0	0	0	1	1	1

Q_2 \ Q_1Q_0	00	01	11	10
0	1	1	1	1
1	1	1	1	1

$T_0 = 1$

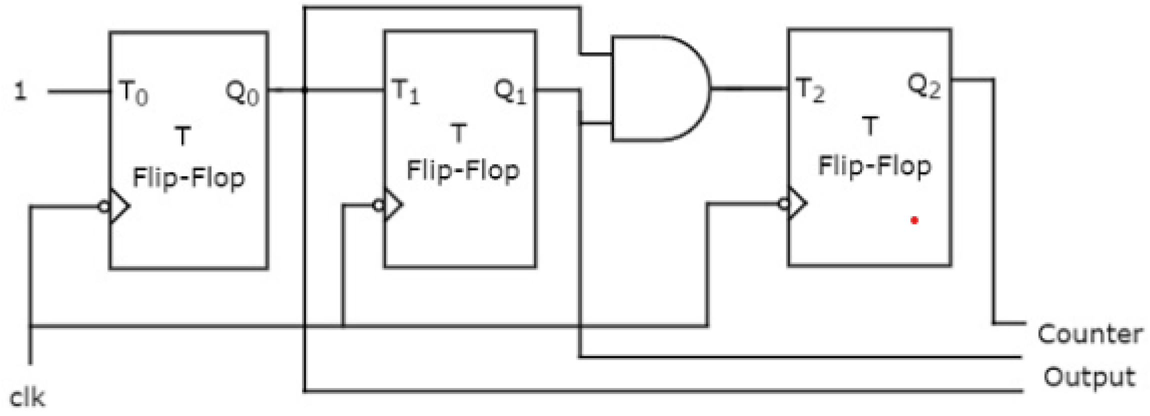
Q_2 \ Q_1Q_0	00	01	11	10
0	0	0	1	0
1	0	0	1	0

$T_2 = Q_1.Q_0$

Q_2 \ Q_1Q_0	00	01	11	10
0	0	1	1	0
1	0	1	1	0

$T_1 = Q_0$

3 bit Synchronous (Parallel) Counters



4 bit Synchronous Decade Couters

0 - 9

0 - 15

	Present State				Next State				Flip flop Input			
Clock pulse	Q_3	Q_2	Q_1	Q_0	Q_{3+}	Q_{2+}	Q_{1+}	Q_{0+}	T_3	T_2	T_1	T_0
Initially	0	0	0	0	0	0	0	1	0	0	0	1
1	0	0	0	1	0	0	1	0	0	0	1	1
2	0	0	1	0	0	0	1	1	0	0	0	1
3	0	0	1	1	0	1	0	0	0	1	1	1
4	0	1	0	0	0	1	0	1	0	0	0	1
5	0	1	0	1	0	1	1	0	0	0	1	1
6	0	1	1	0	0	1	1	1	0	0	0	1
7	0	1	1	1	1	0	0	0	1	1	1	1
8	1	0	0	0	1	0	0	1	0	0	0	1
9	1	0	0	1	0	0	0	0	1	0	0	1

4 bit Synchronous Decade Couters

Q_3Q_2 \ Q_1Q_0	00	01	11	10
00	**0**	**0**	**0**	**0**
01	0	0	1	0
11	0	1	x	x
10	x	x	x	x

$T_3 = Q_2.Q_1.Q_0 + Q_3.Q_0$

Q_3Q_2 \ Q_1Q_0	00	01	11	10
00	**0**	**1**	**1**	**0**
01	0	1	1	0
11	0	0	x	x
10	x	x	x	x

$T_1 = Q'_3.Q_0$

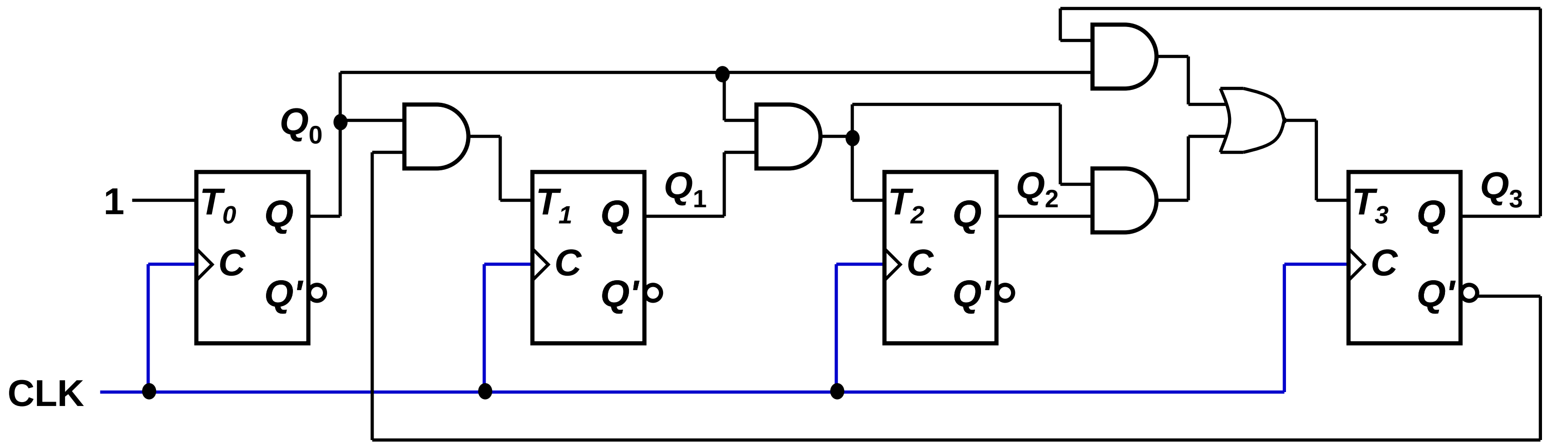
Q_3Q_2 \ Q_1Q_0	00	01	11	10
00	**0**	**0**	**1**	**0**
01	0	0	1	0
11	0	0	x	x
10	x	x	x	x

$T_2 = Q_1.Q_0$

Q_3Q_2 \ Q_1Q_0	00	01	11	10
00	**1**	**1**	**1**	**1**
01	1	1	1	1
11	1	1	x	x
10	x	x	x	x

$T_0 = 1$

4 bit Synchronous Decade Couters



Arbitrary Sequence Counters Design

Step 1: The number of flip-flops required to design counter can be determined by using the equation: $\mathbf{2^n >= N}$, where n is equal to the number of flip-flops and N is the mod number.

Step 2: Drawing the state diagram.

Step 3: make state table

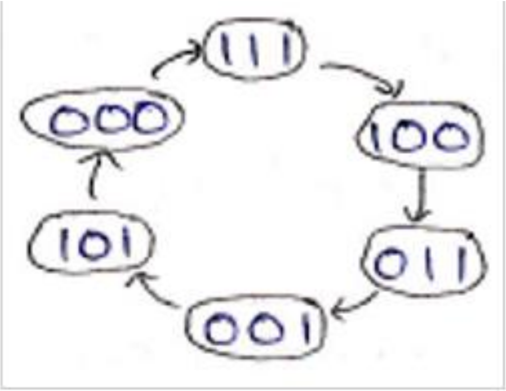
Step 4 Draw circuit

Arbitrary Sequence Counters Design

Design a synchronous counter to count the following sequence 7,4,3,1,5,0,7,4,3,...
Implement the circuit using T-flip-flops.

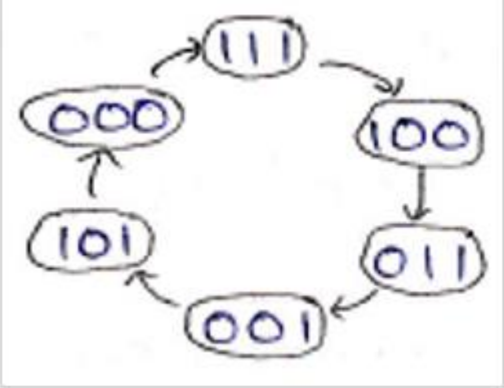
Step 1: Total number of states is 6. the required number of flip-flops is 3.

Step 2: State diagram



Arbitrary Sequence Counters Design

Step 2



Present State			Next State			FF Inputs		
Q_2	Q_1	Q_0	Q_2^+	Q_1^+	Q_0^+	T_2	T_1	T_0
0	0	0	1	1	1	1	1	1
0	0	1	1	0	1	1	0	0
0	1	0	X	X	X	X	X	X
0	1	1	0	0	1	0	1	0
1	0	0	0	1	1	1	1	1
1	0	1	0	0	0	1	0	1
1	1	0	X	X	X	X	X	X
1	1	1	1	0	0	0	1	1

Solving for T_2 →

T_2: $Q_2 \backslash Q_1Q_0$	00	01	11	10
0	1	1		X
1	1	1		X

$$\boxed{T_2 = \overline{Q_1}}$$

Solving for T_1 →

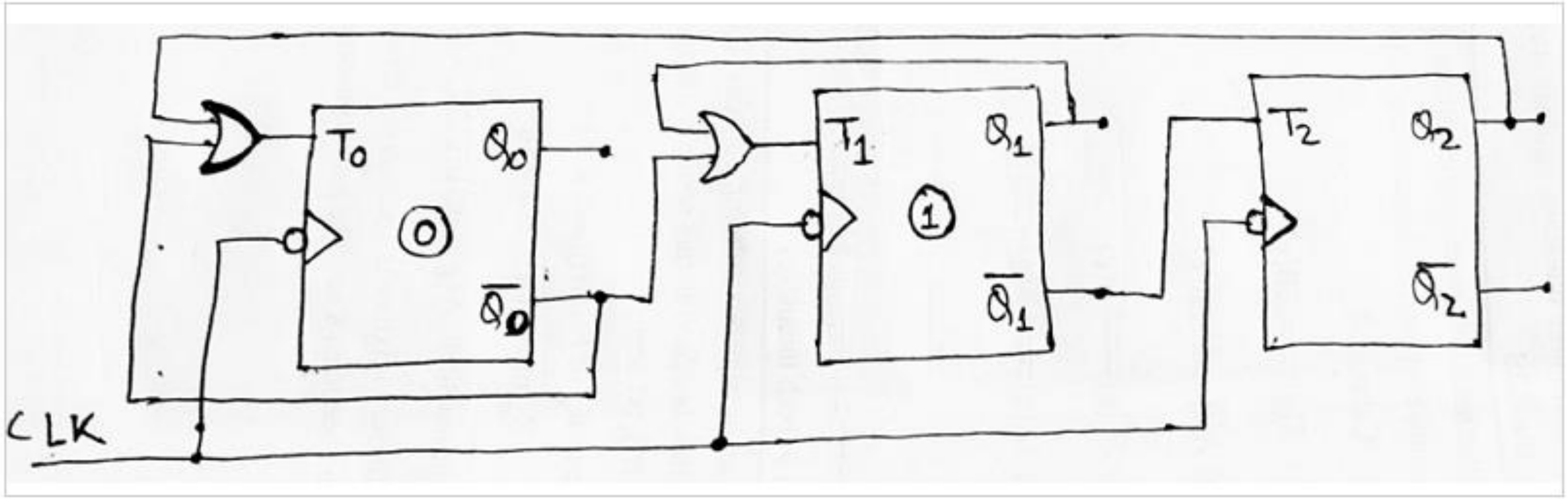
T_1: $Q_2 \backslash Q_1Q_0$	00	01	11	10
0	1		1	X
1	1		1	X

$$\boxed{T_1 = Q_1 + Q_0'}$$

T_0: $Q_2 \backslash Q_1Q_0$	00	01	11	10
0	1			X
1	1	1	1	X

$$T_0 = Q_2 + \overline{Q_0}$$

Arbitrary Sequence Counters Design

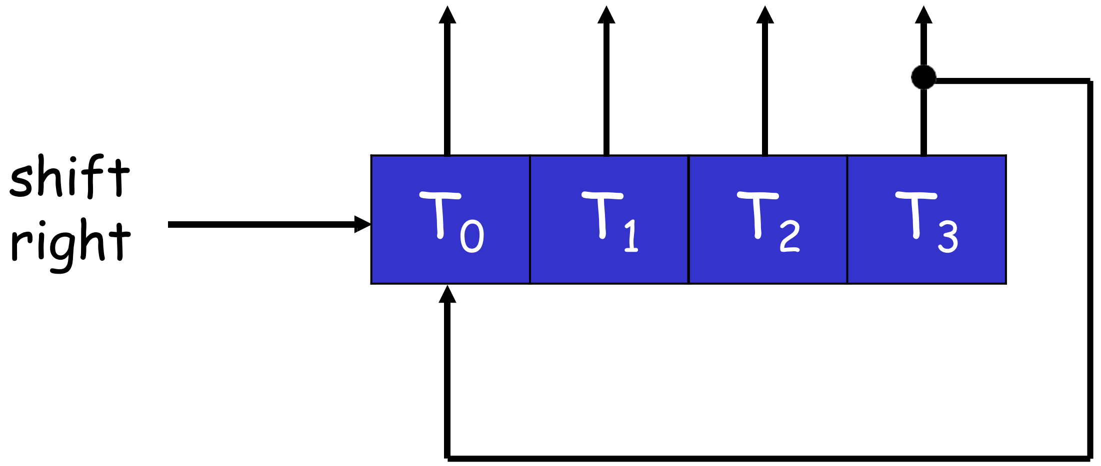


Ring counter

- Ring Counter
  - Timing signals control the sequence of operations in a digital system
  - A ring counter is a circular shift register with only one flip-flop being set at any particular time, all others are cleared.

almost the same as the shift counter.

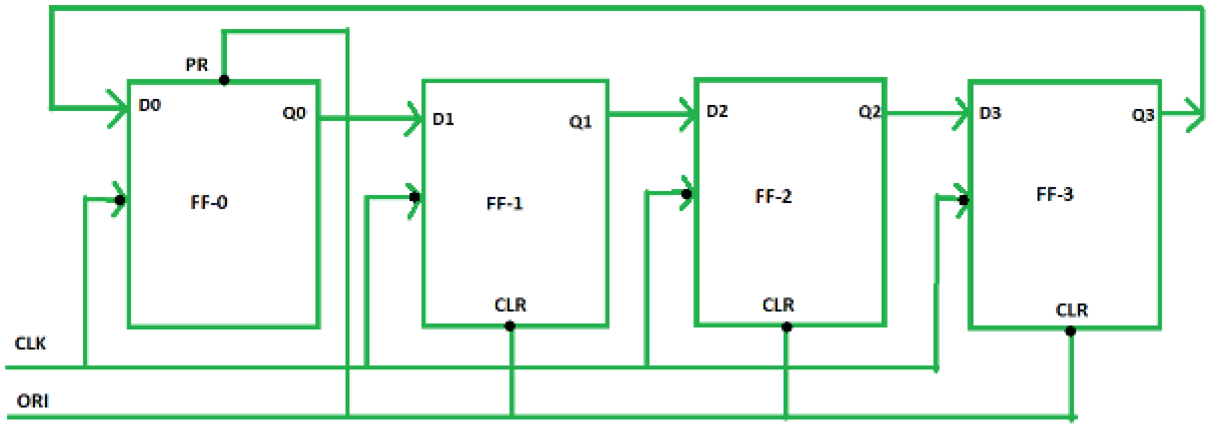
The only change is that the output of the last flip-flop is connected to the input of the first flip-flop in the case of the ring counter



initial value
1000

4 bit Ring counter

clock pulse (CLK) is applied to all the flip-flops simultaneously. Therefore, it is a Synchronous Counter.



Ring Counter

Overriding input (ORI) for each flip-flop. Preset (PR) and Clear (CLR) are used as ORI.

When PR is 0, then the output is 1. And when CLR is 0, then the output is 0. Both PR and CLR are active low signal that always works in value 0.

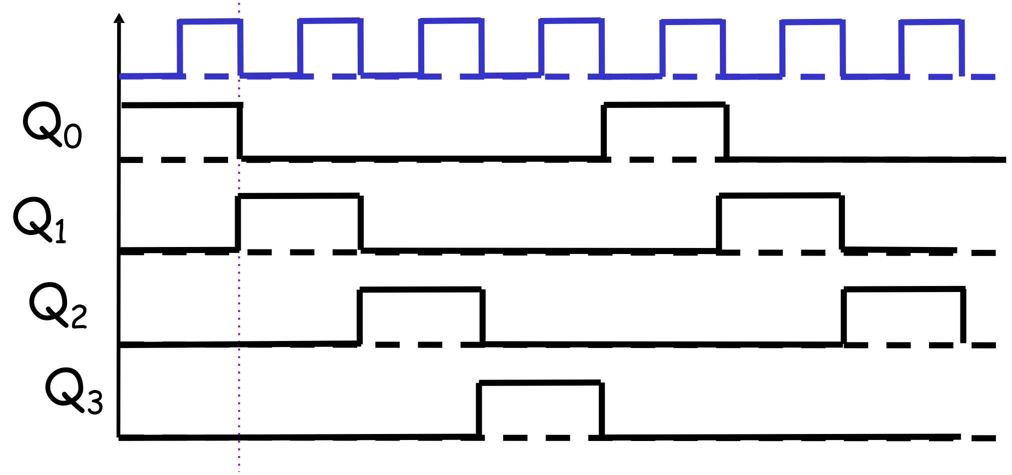
ORI is connected to Preset (PR) in FF-0 and it is connected to Clear (CLR) in FF-1, FF-2, and FF-3. Thus, output Q = 1 is generated at FF-0, and the rest of the flip-flop generates output Q = 0. This output Q = 1 at FF-0 is known as Pre-set 1 which is used to form the ring in the Ring Counter.

4 bit Ring counter

PRESETED 1

ORI	CLK	Q0	Q1	Q2	Q3
low	X	1	0	0	0
high	low	0	1	0	0
high	low	0	0	1	0
high	low	0	0	0	1
high	low	1	0	0	0

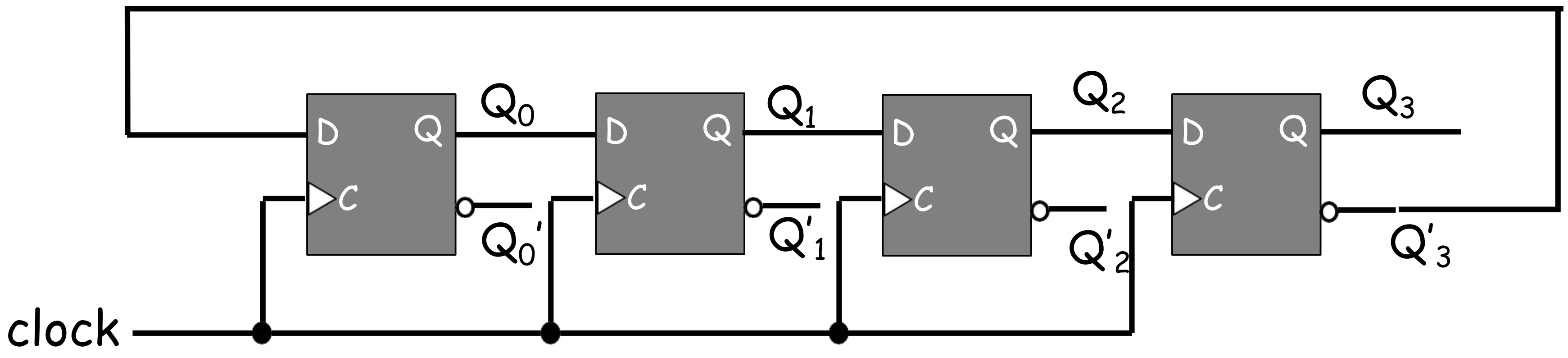
- Sequence of timing signals



at each clock pulse, the preseted 1 is shifted to the next flip-flop and thus forms a Ring. From the above table, we can say that there are 4 states in a 4-bit Ring Counter.

Johnson counter

- A k-bit ring counter can generate k distinguishable states
- The number of states can be doubled if the shift register is connected as a <u>switch-tail</u> ring counter



Johnson counter

Table 1: Truth table of 4-bit Johnson Counter

State	Q0	Q1	Q2	Q3
0	0	0	0	0
1	1	0	0	0
2	1	1	0	0
3	1	1	1	0
4	1	1	1	1
5	0	1	1	1
6	0	0	1	1
7	0	0	0	1
8	0	0	0	0